

Integration

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Lec1

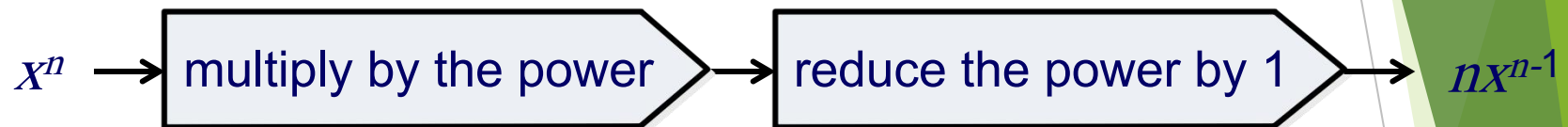
Reversing the process of differentiation

- **Reversing the process of differentiation**
- Indefinite integration
- Rewriting expressions before integrating
- Finding the constant of integration given a point
- Examination-style questions

Reversing the process of differentiation

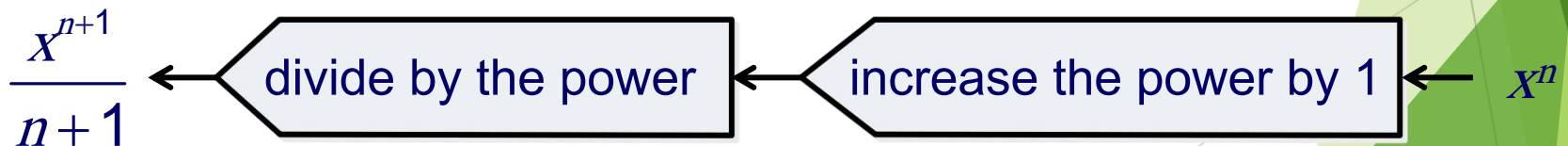
To differentiate $y = x^n$ with respect to x we multiply by the power and reduce the power by one.

We can write this process as follows:



Suppose we are given the derivative $\frac{dy}{dx} = x^n$ and asked to find y in terms of x .

Reversing the process of differentiation given above would give



Reversing the process of differentiation

For example: If $\frac{dy}{dx} = 6x^2$ find y in terms of x

Adding 1 to the power and dividing by the new power gives:

$$\begin{aligned} y &= \frac{6}{3} x^3 \\ &= 2x^3 \end{aligned}$$

This is not the complete solution, however, because if we differentiated $y = 2x^3 + 1$,

$$\text{or } y = 2x^3 - 3,$$

$$\text{or } y = 2x^3 + \text{any constant}$$

we would also get $\frac{dy}{dx} = 6x^2$

We therefore have to write $y = 2x^3 + c$.

Reversing the process of differentiation

We can't find the value of c without being given further information. It is an **arbitrary constant**.

The process of finding a function given its derivative is called **integration**.

We call c **the constant of integration**.

The integral of $6x^2$ with respect to x is written as:

$$\int 6x^2 dx = 2x^3 + c$$

This is called an **indefinite integral** because we don't know the value of c .

Indefinite integration

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Indefinite integration

In general:

$$\int ax^n dx = \frac{ax^{n+1}}{n+1} + c \text{ for all } n \neq -1$$

$\int ax^n dx$ can also be written as $a \int x^n dx$, where a is a constant.

For example:

$$\text{Find } a) \int 12x^5 dx \quad b) \int 8 dx \quad c) \int \frac{1}{2} x^3 dx$$

$$\begin{aligned} a) \int 12x^5 dx &= \frac{12x^6}{6} + c & b) \int 8 dx &= 8x + c & c) \int \frac{x^3}{2} dx &= \frac{1}{2} \times \frac{x^4}{4} + c \\ &= 2x^6 + c & & & &= \frac{x^4}{8} + c \end{aligned}$$

Integration of polynomials

When we differentiated polynomials we differentiated each term at a time.

The same can be done when polynomials are integrated.

For example:

$$\text{Find } \int (8x^3 + 5x^2 - 6x + 2) dx$$

$$\begin{aligned}\int (8x^3 + 5x^2 - 6x + 2) dx &= \int 8x^3 dx + \int 5x^2 dx - \int 6x dx + \int 2 dx \\ &= \frac{8x^4}{4} + \frac{5x^3}{3} - \frac{6x^2}{2} + 2x + c \\ &= 2x^4 + \frac{5}{3}x^3 - 3x^2 + 2x + c\end{aligned}$$

In general:

$$\int (f(x) + g(x)) dx = \int f(x) dx + \int g(x) dx$$

Integration of ax^n for all rational n

The rule

$$\int ax^n dx = \frac{ax^{n+1}}{n+1} + c$$

can be applied to all negative or fractional values of n except $n = -1$.

For example:

$$\text{Find } \int \frac{2}{x^2} dx$$

Start by writing this as

$$\begin{aligned} & \int 2x^{-2} dx \\ &= -2x^{-1} + c \\ &= -\frac{2}{x} + c \end{aligned}$$

Integration of ax^n for all rational n

$$\text{Find } \int \left(3\sqrt{x} + \frac{2}{\sqrt{x}} \right) dx$$

This can be written as $\int (3x^{\frac{1}{2}} + 2x^{-\frac{1}{2}}) dx$

$$= \frac{3x^{\frac{3}{2}}}{\frac{3}{2}} + \frac{2x^{\frac{1}{2}}}{\frac{1}{2}} + c$$

Remember, to divide by $\frac{3}{2}$ we multiply by $\frac{2}{3}$ and to divide by $\frac{1}{2}$ we multiply by 2. So

$$\frac{3x^{\frac{3}{2}}}{\frac{3}{2}} + \frac{2x^{\frac{1}{2}}}{\frac{1}{2}} + c = 2x^{\frac{3}{2}} + 4x^{\frac{1}{2}} + c$$

Rewriting expressions before integrating

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Rewriting expressions before integrating

As with differentiation, some expressions will need to be rewritten as separate terms of the form ax^n before integrating.
For example:

$$\text{Find } \int (3 - 2x)^2 dx$$

Expanding the brackets gives:

$$\begin{aligned}\int (3 - 2x)^2 dx &= \int (9 - 12x + 4x^2) dx \\ &= 9x - \frac{12x^2}{2} + \frac{4x^3}{3} + c \\ &= 9x - 6x^2 + \frac{4x^3}{3} + c\end{aligned}$$

Rewriting expressions before integrating

$$\text{Find } \int \frac{4x^3 + 5x^2 - 2}{x^2} dx$$

We can write this as $\int \left(\frac{4x^3}{x^2} + \frac{5x^2}{x^2} - \frac{2}{x^2} \right) dx$

$$= \int (4x + 5 - 2x^{-2}) dx$$

$$= \frac{4x^2}{2} + 5x - \frac{2x^{-1}}{-1} + c$$

$$= 2x^2 + 5x + 2x^{-1} + c$$

Finding the constant of integration

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Finding the constant of integration given a point

Suppose we know the gradient function of a curve and we want to find its equation.

We can do this by integration if we are also given a point on the curve.

This is because we can use the coordinates of the point to find the constant of integration. For example:

A curve $y = f(x)$ passes through the point $(2, 9)$.

Given that $\frac{dy}{dx} = 8x^3 - 10x$ find the equation of the curve.

$$\begin{aligned}\text{If } \frac{dy}{dx} = 8x^3 - 10x \text{ then } y &= \int (8x^3 - 10x) dx \\ &= \frac{8x^4}{4} - \frac{10x^2}{2} + c \\ &= 2x^4 - 5x^2 + c\end{aligned}$$

Finding the constant of integration given a point

The curve passes through the point (2, 9) and so we can substitute $x = 2$ and $y = 9$ into the equation of the curve to find the value of c .

$$y = 2x^4 - 5x^2 + c$$

$$9 = 2(2)^4 - 5(2)^2 + c$$

$$9 = 32 - 20 + c$$

$$9 = 12 + c$$

$$c = -3$$

So the equation of the curve is $y = 2x^4 - 5x^2 - 3$.

Examination-style questions

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Examination-style question

A curve has gradient function $6x^2 - 7$. Given that the curve passes through the point with coordinates $(2, 5)$ find the equation of the curve.

$$f'(x) = 6x^2 - 7$$

$$f(x) = 2x^3 - 7x + c$$

Substituting $x = 2$ and $y = 5$ into this equation gives:

$$5 = 16 - 14 + c$$

$$c = 3$$

So the equation of the curve is $y = 2x^3 - 7x + 3$.



Basic Formulae

$$(i) \int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$$

$$(ii) \int e^{ax} dx = \frac{e^{ax}}{a} + C$$

$$(iii) \int a^x dx = \frac{a^x}{\log a} + C$$

$$(iv) \int \sin x dx = -\cos x + C$$

$$(v) \int \cos x dx = \sin x + C$$

$$(vi) \int \tan x dx = -\log|\cos x| + C = \log|\sec x| + C$$

$$(vii) \int \cot x dx = \log|\sin x| + C = -\log|\operatorname{cosec} x| + C$$

$$(viii) \int \sec x dx = \log|\sec x + \tan x| + C = \log\left|\tan\left(\frac{\pi}{4} + \frac{x}{2}\right)\right| + C$$

$$(ix) \int \operatorname{cosec} x dx = \log|\operatorname{cosec} x - \cot x| + C = \log\left|\tan\frac{x}{2}\right| + C$$

$$(x) \int \sec x \tan x dx = \sec x + C$$

$$(xi) \int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$$

Basic Formulae

$$(xii) \int \sec^2 x dx = \tan x + C$$

$$(xiv) \int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + C$$

$$(xvi) \int \frac{1}{1+x^2} dx = \tan^{-1} x + C$$

$$(xviii) \int \frac{1}{x\sqrt{x^2-1}} dx = \sec^{-1} x + C$$

$$(xx) \int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1} \frac{x}{a} + C$$

$$(xxii) \int \frac{dx}{\sqrt{x^2+a^2}} = \log |x + \sqrt{x^2+a^2}| + C$$

$$(xiii) \int \operatorname{cosec}^2 x dx = -\cot x + C$$

$$(xv) \int \frac{-1}{\sqrt{1-x^2}} dx = \cos^{-1} x + C$$

$$(xvii) \int \frac{-1}{1+x^2} dx = \cot^{-1} x + C$$

$$(xix) \int \frac{-1}{x\sqrt{x^2-1}} dx = \operatorname{cosec}^{-1} x + C$$

$$(xxi) \int \frac{dx}{\sqrt{x^2-a^2}} = \log |x + \sqrt{x^2-a^2}| + C$$

$$(xxiii) \int \frac{1}{a^2+x^2} dx = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$$

$$(xxiv) \int \frac{1}{a^2 - x^2} dx = \frac{1}{2a} \log \left| \frac{a+x}{a-x} \right| + C$$

$$(xxv) \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + C$$

$$(xxvi) \int \sqrt{x^2 - a^2} dx = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \log \left| x + \sqrt{x^2 - a^2} \right| + C$$

$$(xxvii) \int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + C$$

$$(xxviii) \int \sqrt{x^2 + a^2} dx = \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \log \left| x + \sqrt{x^2 + a^2} \right| + C$$

$$(xxix) \int (ax+b)^n dx = \frac{1}{a} \frac{(ax+b)^{n+1}}{n+1} + C, n \neq -1$$

$$(xxx) \int e^x [f(x) + f'(x)] dx = f(x) e^x + C$$

The end

